COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE 010- Data Structures and Algorithms	**SECTION:**	CPE21S4
DATE PERFORMED:	**10/1/2026**	**DATE SUBMITTED:**	**10/1/2026**
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SEATWORK NO. 10.1 **[Graph]**

1. PROCEDURE OUTPUT

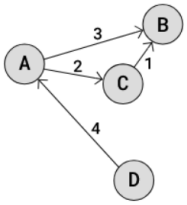
What procedures did you perform?

General Instructions:

1. Define the following: (10)

   Graph
   Vertex
   Nodes
   Weight
   Path
   Directed graph
   Connected graph
   Directed cyclic
   Adjacency matrix
   Adjacency List

2. Identify the parts of the following graph. (10)



3. Plot the adjacency matrix of the graph above. (10 pts)

4. Create a program in C++ program to create a graph that looks like in the above figure.

5. Differentiate a BFS vs DFS types of search.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. Define the following: (10)

A.Graph – A group of points and connections.

B.Vertex – A point in a graph.

C.Nodes – The points in a graph.

D.Weight – A number that shows the cost or distance.

E.Path – A way to go from one point to another.

F.Directed Graph – A graph where connections have a direction.

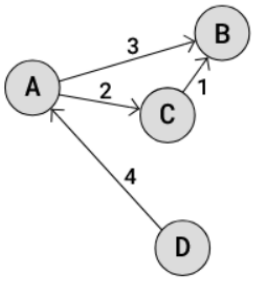
G.Connected Graph – A graph where all points are connected.

H.Directed Cyclic – A directed graph that can go back to the starting point.

I.Adjacency Matrix – A table that shows the connections between points.

J.Adjacency List – A list that shows which points are connected.

2. Identify the parts of the following graph. (10)



1.)Vertex

2.)Edge

3.)Weight

4.)Vertex

3. Plot the adjacency matrix of the graph above. (10 pts)

	A	B	C	D
A		3	2	
B				
C		1		
D	4			

4. Create a program in C++ program to create a graph that looks like in the above figure.

```
#include <iostream>

#include <vector>

#include <map>

#include <iomanip>


using namespace std;


struct Edge {

    char target;

    int weight;

};


int main() {
```

```
    // 1. Adjacency List Representation
    map<char, vector<Edge>> adjList = {
        {'A', {{'B', 3}, {'C', 2}}},
        {'B', {}},
        {'C', {{'B', 1}}},
        {'D', {{'A', 4}}}
    };

    cout << "=== Adjacency List Representation ===" << endl;
    for (const auto& [node, edges] : adjList) {
        cout << node << " -> ";
        if (edges.empty()) {
            cout << "None";
        } else {
            for (const auto& edge : edges) {
                cout << "(" << edge.target << ", w:" << edge.weight << ")
";
            }
        }
        cout << endl;
    }

    vector<char> nodes = {'A', 'B', 'C', 'D'};
    int adjMatrix[4][4] = {
```

```
        {0, 3, 2, 0}, // Row A
        {0, 0, 0, 0}, // Row B
        {0, 1, 0, 0}, // Row C
        {4, 0, 0, 0}  // Row D
    };

    cout << "\n=== Adjacency Matrix Representation ===" << endl;
    cout << "    ";
    for (char node : nodes) cout << setw(4) << node;
    cout << "\n--------------------" << endl;

    for (int i = 0; i < 4; i++) {
        cout << nodes[i] << " | ";
        for (int j = 0; j < 4; j++) {
            cout << setw(4) << adjMatrix[i][j];
        }
        cout << endl;
    }

    return 0;
}
```

5. Differentiate a BFS vs DFS types of search.

BFS (Breadth-First Search):

How it works: You check *every* room on Floor 1, then every room on Floor 2, then Floor 3.

Analogy: Like ripples expanding outward when a stone drops in water.

Best for: Finding the **shortest route** or the closest match.

DFS (Depth-First Search) :

How it works: You pick one hallway, follow it all the way to the very end (dead end), turn around, and try the next hallway.

Analogy: Like exploring a maze.

Best for: Finding **any solution/exit** and using less memory.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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